INCH-POUND
MIL-DTL-16377/21B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/21A(SH)
11 April 1983

DETAIL SPECIFICATION SHEET

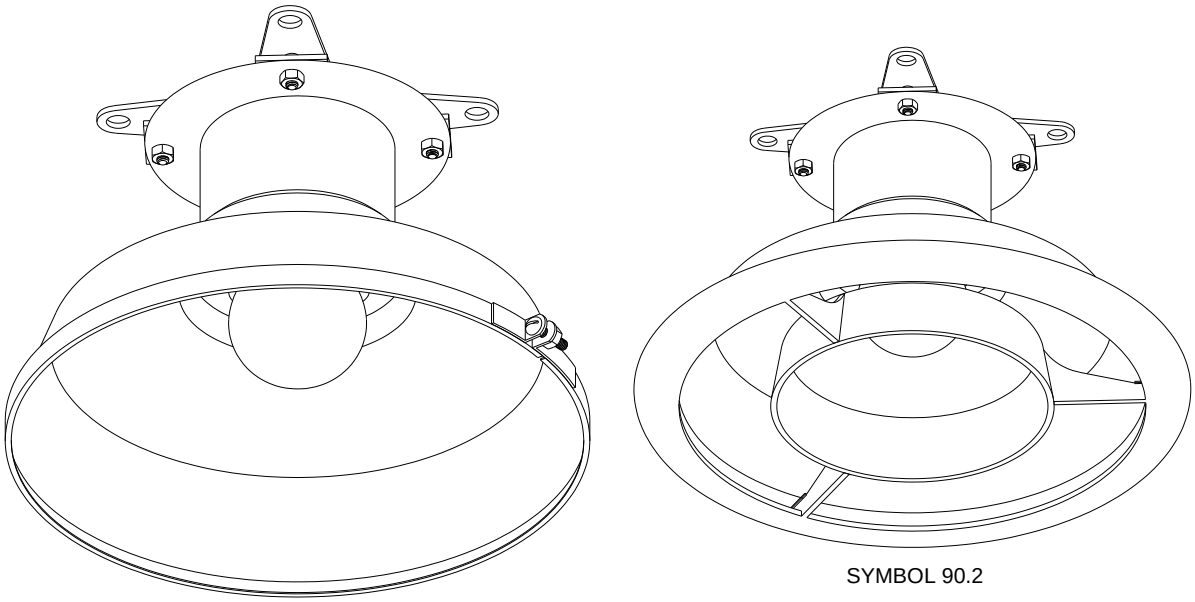
# FIXTURES, LIGHTING; INCANDESCENT AND SOLID-STATE EQUIVALENT, GENERAL LIGHTING, 100 WATTS, 120 VOLTS, DRIPPROOF 15 DEGREES, SYMBOLS 89, 90.2, 89SSL, AND 90.2SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 9 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.



SYMBOL 89

SYMBOL 90.2

FIGURE 1. Typical view of symbols 89 and 90.2.



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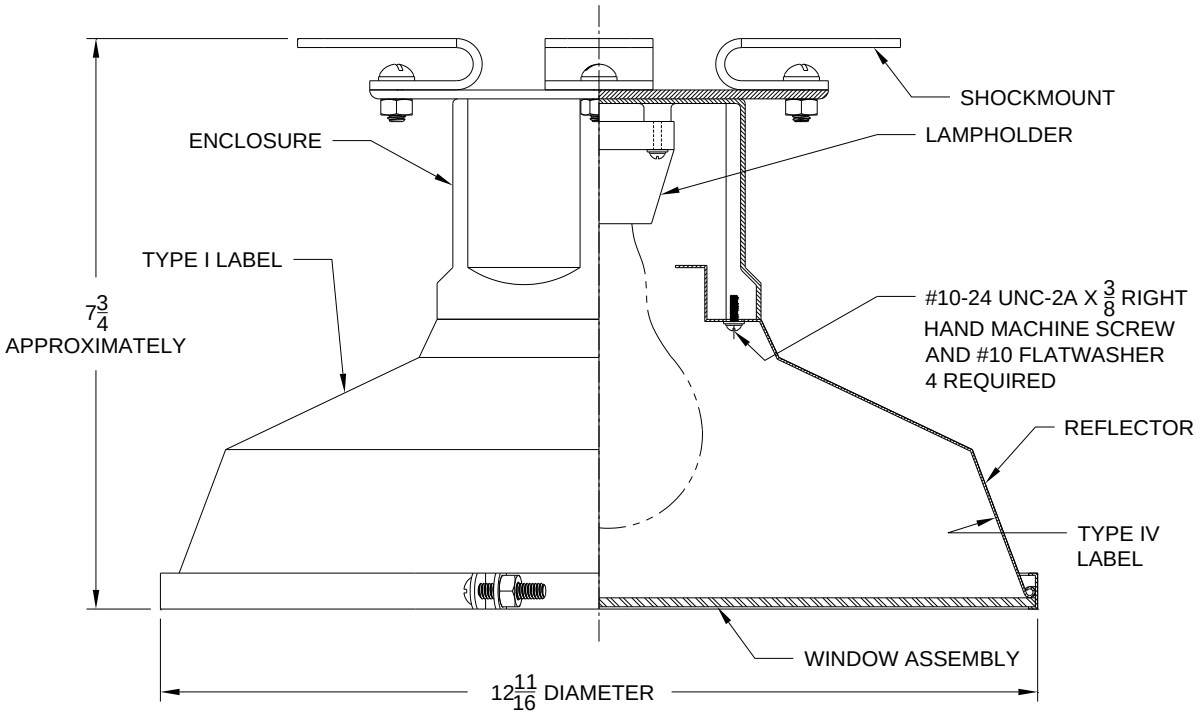



FIGURE 2. Fixture assembly, symbol 89.

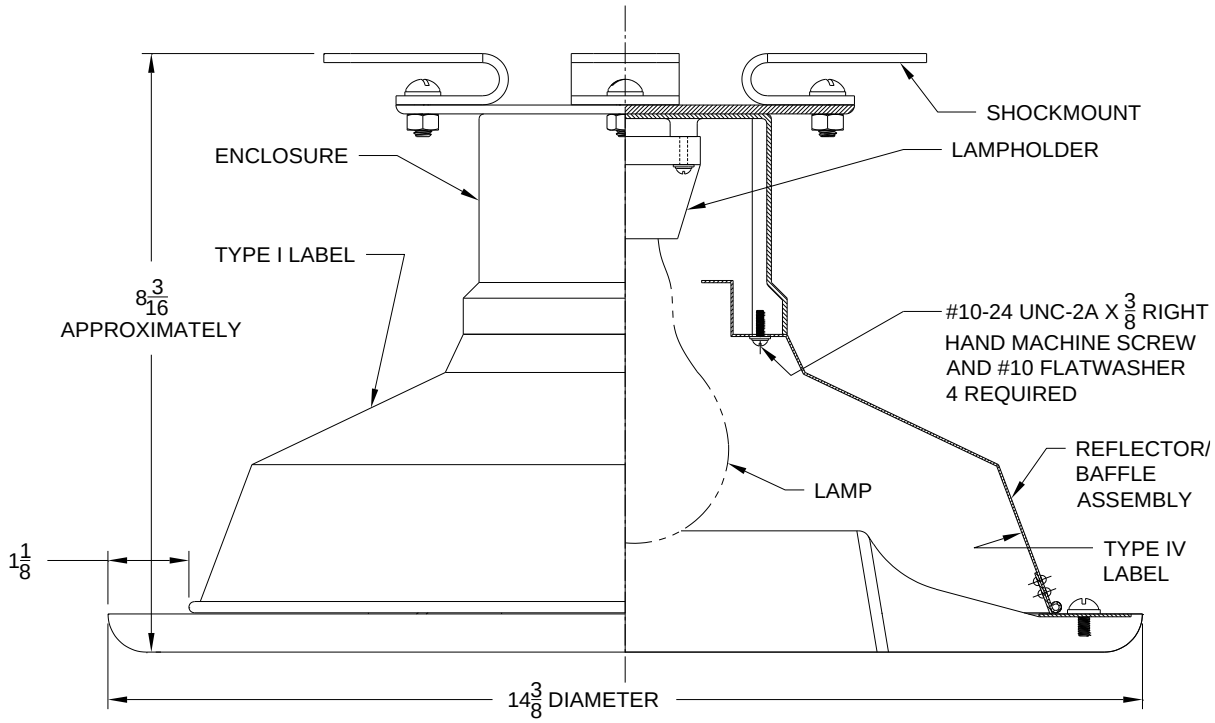



FIGURE 3. Fixture assembly, symbol 90.2.



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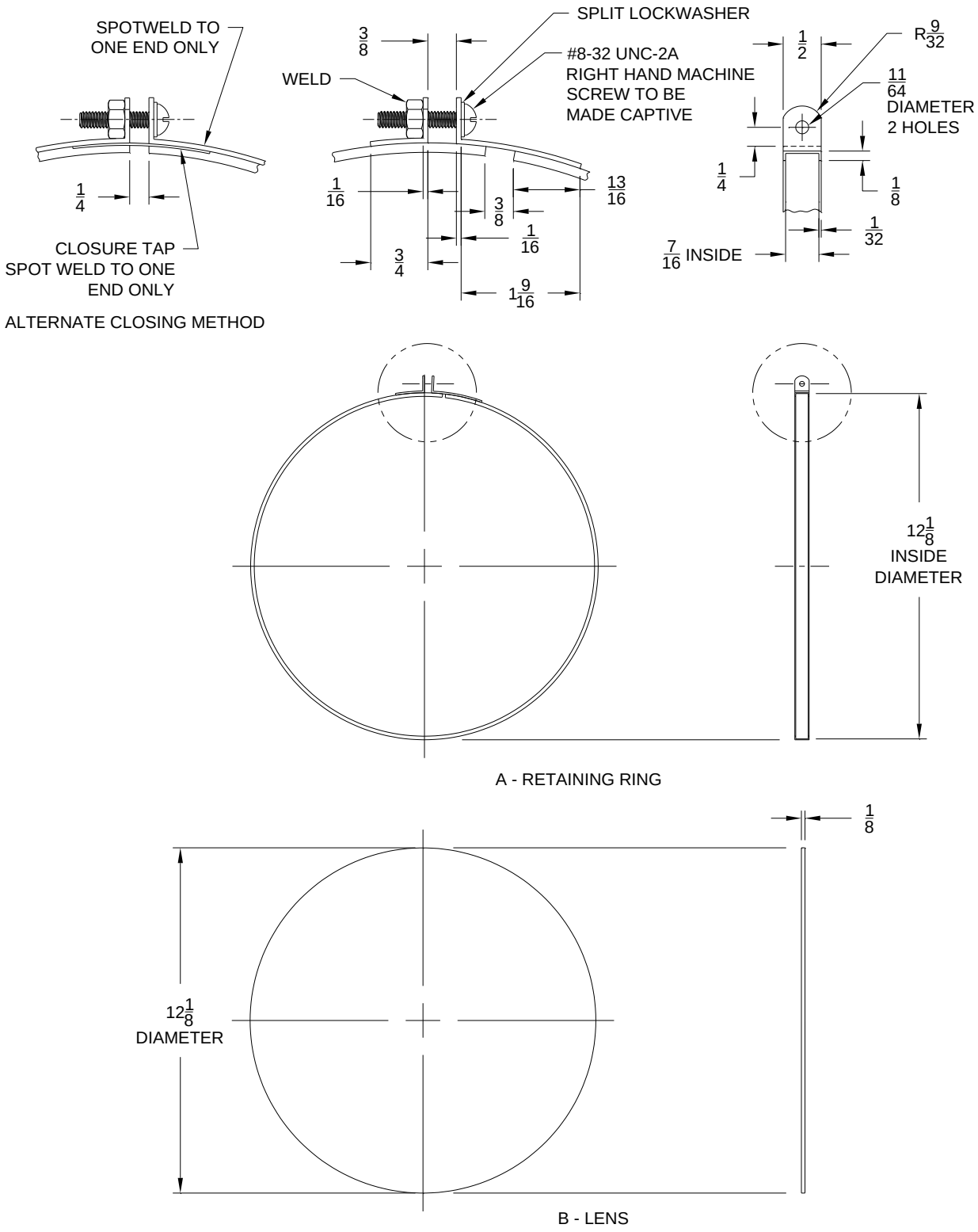



FIGURE 4. Window assembly.



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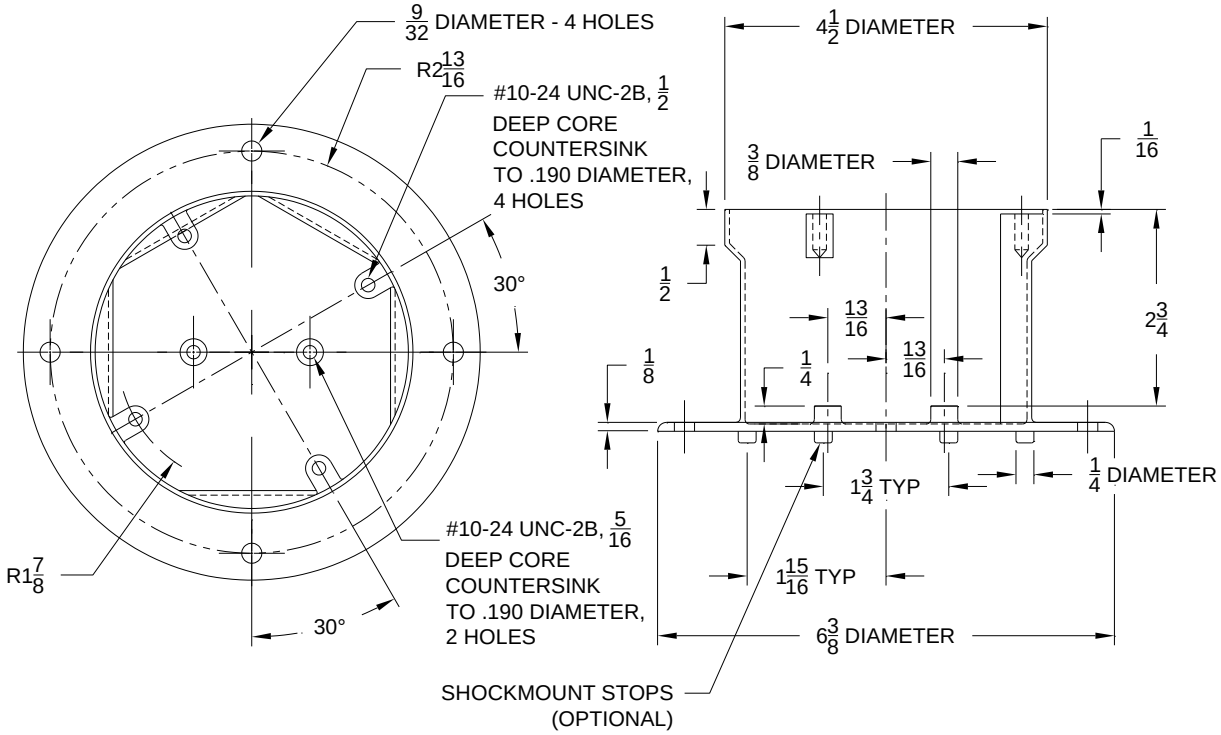



FIGURE 5. Enclosure.



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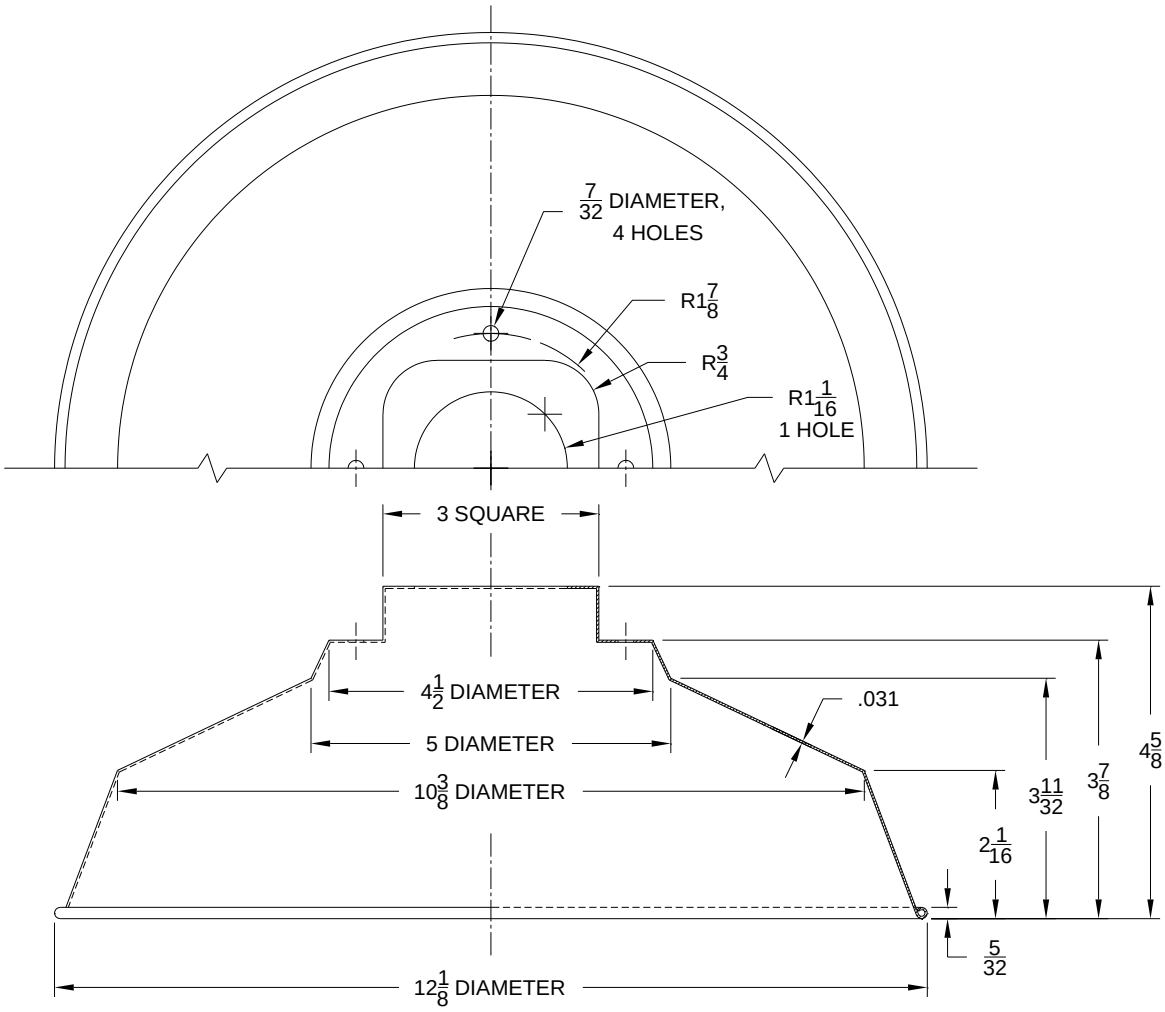



FIGURE 6. Reflector.



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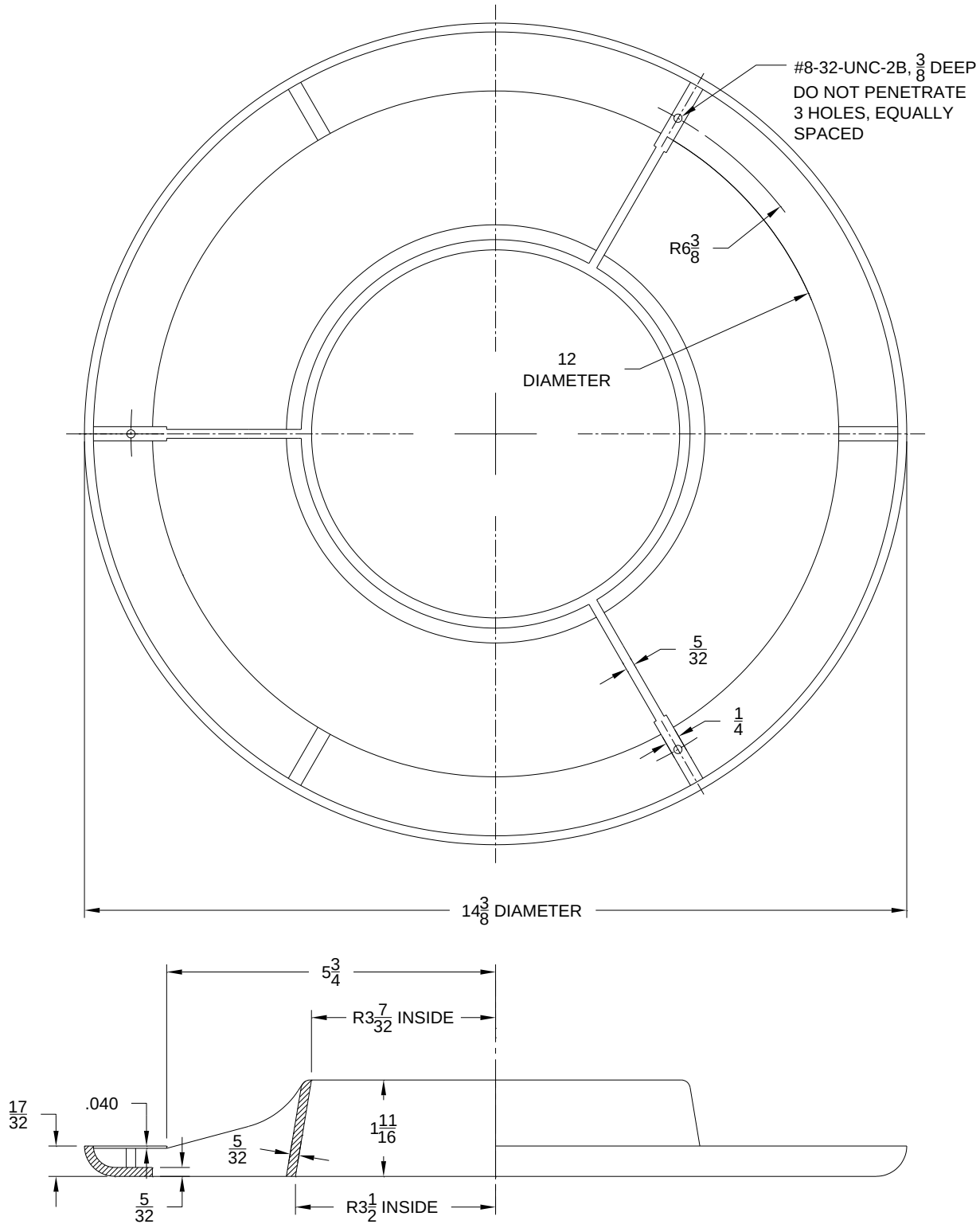



FIGURE 7. Baffle.



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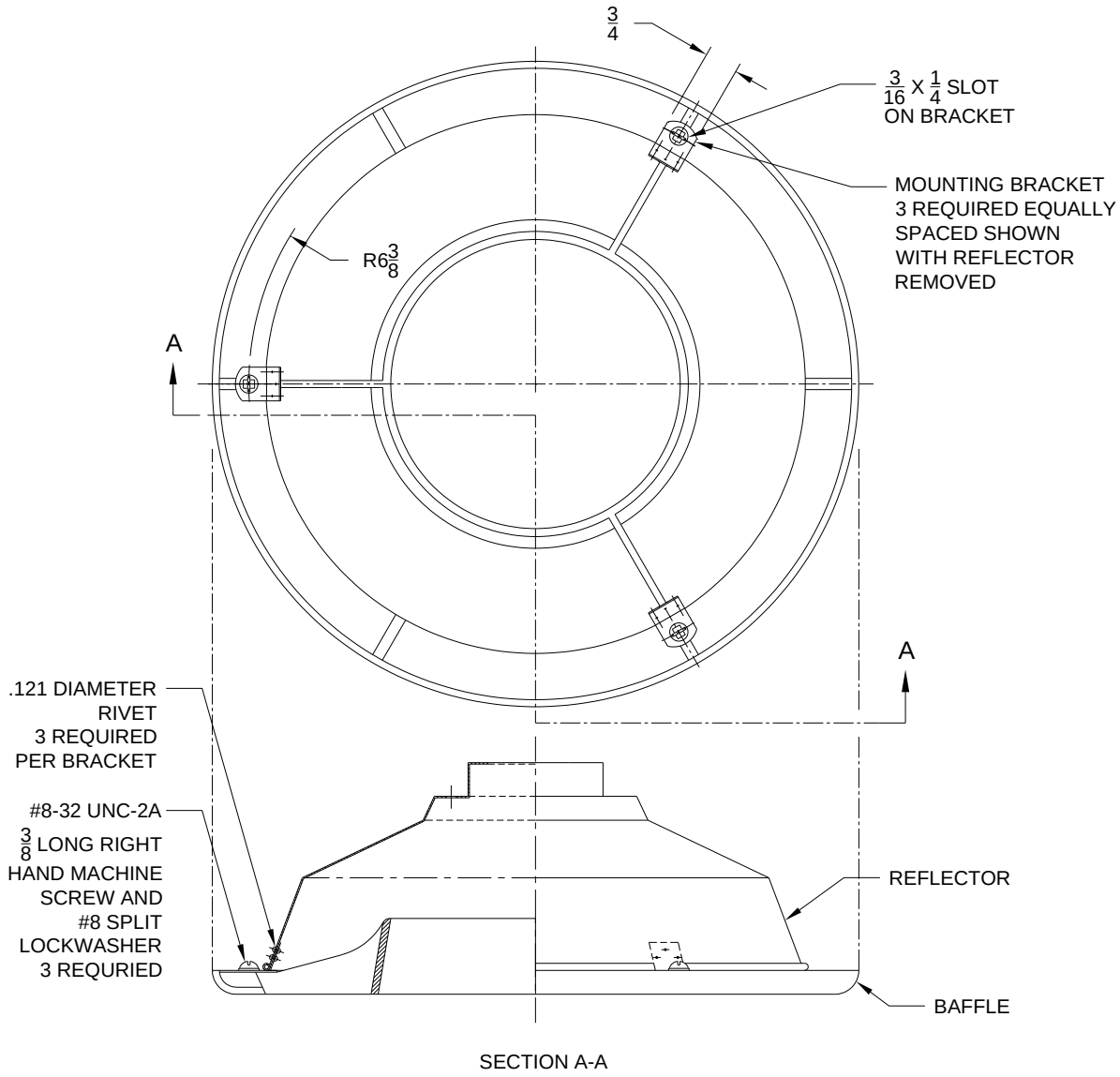
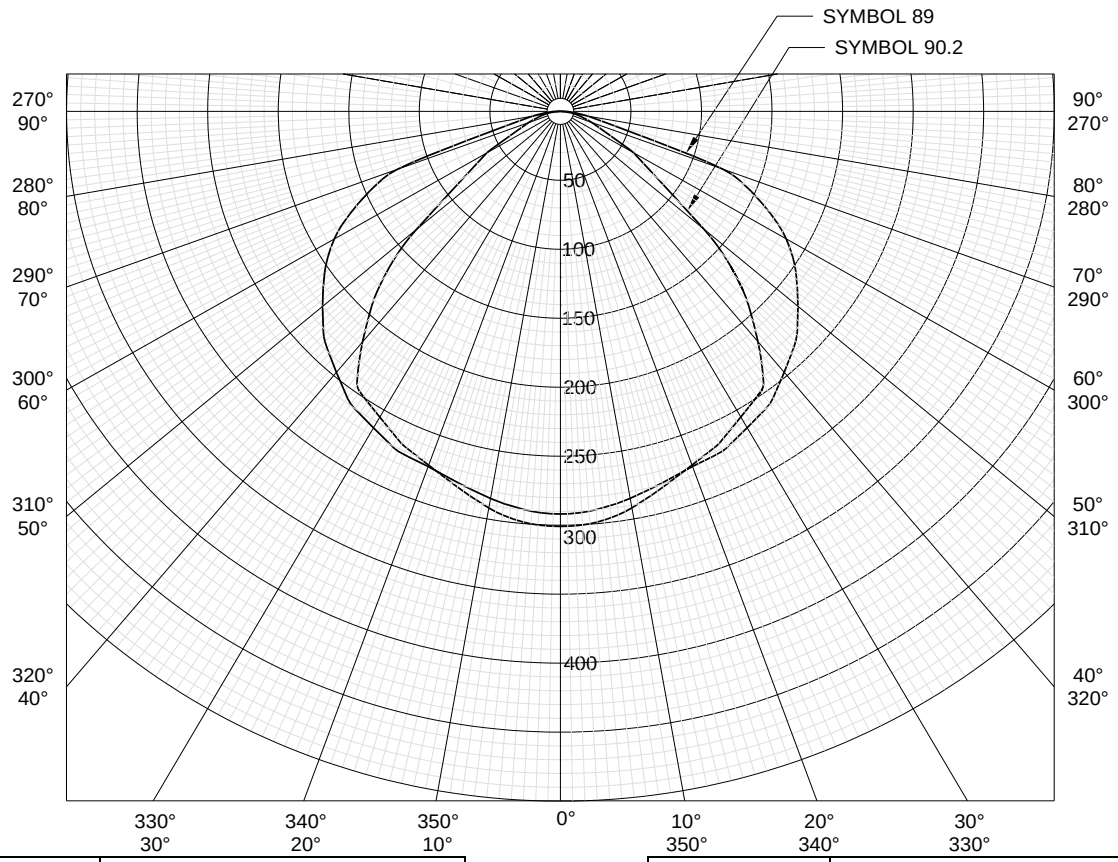



FIGURE 8. Reflector/baffle assembly.



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| ANGLE | CANDLEPOWER | |
|---|---|---|
| | SYM 89 | SYM 90.2 |
| 0 | 292 | 301 |
| 5 | 290 | 299 |
| 10 | 285 | 292 |
| 15 | 279 | 282 |
| 20 | 274 | 273 |
| 25 | 272 | 266 |
| 30 | 265 | 256 |
| 35 | 259 | 248 |
| 40 | 247 | 218 |
| 45 | 236 | 183 |
| 50 | 220 | 133 |
| 55 | 204 | 83 |
| 60 | 185 | 60 |

| ANGLE | CANDLEPOWER | |
|---|---|---|
| | SYM 89 | SYM 90.2 |
| 65 | 158 | 35 |
| 70 | 123 | 24 |
| 75 | 38 | 16 |
| 80 | 20 | 10 |
| 85 | 10 | 4 |
| 90 | 0 | 0 |
| 95 | --- | --- |
| 100 | --- | --- |
| 105 | --- | --- |
| 110 | --- | --- |
| 115 | --- | --- |
| 120 | --- | --- |
| 125 | --- | --- |

Photometric test data:

Distance: 10 feet

Voltage: 120 volts

Ambient temperature: 76 °F

Lamp: 100 watts, rough service conforming to W-L-101/77, furnished by laboratory for test, shall be used

FIGURE 9. Candlepower distribution curves.



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## INCANDESCENT REQUIREMENTS

CLASSIFICATION

Type: II
Class: 2

SYMBOL NUMBERS

89: Complete lighting fixture assembly with reflector and window assembly.
90.2: Complete lighting fixture assembly with reflector and baffle assembly.

DESIGN

| | |
|---|---|
| Electrical rating: | 100 watts, 120 volts. |
| Dimensions and configuration: | See figures 1 through 8. |
| Mounting dimensions: | See figure 28 of MIL-DTL-16377. |
| Photometric data: | See figure 9. |
| Enclosure assembly: | Material for enclosure and reflector shall be aluminum. |
| Window assembly (symbol 89 only): | See figure 4. Material for lens shall be polycarbonate, clear, with light transmittance exceeding 80 percent, and shall conform to the property values of ASTM D3935. A retaining ring as shown on figure 4 shall be provided for attaching the window to the reflector rim. All metal parts of the retaining ring shall be steel, zinc plated. |
| Lampholder: | UL 496, Supplement SB, double-contact, medium-screw. |
| Baffle assembly (symbol 90.2 only): | Aluminum, cast or fabricated. |
| Shock mounts: | MIL-F-16377/41, part number M16377/41-009. |
| Enclosure: | Dripproof. |
| Finish: | All surfaces, except shock mounts and window assembly, shall receive a baked white enamel finish. |
| Weight: | |
| Symbol 89: | 2 pounds, 14 ounces maximum. |
| Symbol 90.2: | 3 pounds, 4 ounces maximum. |
| Lamp: | These fixtures shall be designed to accommodate a 100-watt, 120-volt, A-21 bulb, inside frosted, medium-screw base, rough-service lamp conforming to W-L-101/77. (Lamp not furnished.) |
| Designation and marking: | The following labels shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 2 and 3:<br>(a) One type I.<br>(b) One type IV. Apply one type IV label (as shown on figures 2 and 3) with the following information in 3/16-inch letters: "USE LAMP NSN6420-00-246-5052 ONLY." |

## SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 2

SYMBOL NUMBERS

89SSL: Complete SSL fixture assembly.
90.2SSL: Complete SSL fixture assembly.



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DESIGN

| | |
|---|---|
| Electrical rating: | 50 watts maximum, 120 volts, 60 hertz. |
| Dimensions and configuration: | See figures 2 and 3. |
| Photometric data: | See figure 9 and 3.7.1 of MIL-DTL-16377. |
| Chromaticity: | See 3.7.4 of MIL-DTL-16377. |
| Fail-safe circuit: | See 3.6.17 of MIL-DTL-16377. |
| Mounting dimensions: | |
| Symbol 89SSL: | See figure 2 and figure 28 of MIL-DTL-16377. |
| Symbol 90.2SSL: | See figure 3 and figure 28 of MIL-DTL-16377. |
| Enclosure: | Dripproof. See 3.5.11 of MIL-DTL-16377. |
| Weight: | |
| Symbol 89SSL: | 2 pounds, 4 ounces maximum. |
| Symbol 90.2SSL: | 2 pounds, 0 ounces maximum. |
| Designation and marking: | One type I label shall be furnished in accordance with MIL-DTL-16377 and applied as shown on figures 2 and 3. |

## EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

| | |
|---|---|
| First article inspection: | |
| Complete fixtures: | See table I for type II and table III for type III. |
| Repair parts: | See table II. |
| Conformance inspection: | |
| Comparison inspection: | Same as the first article inspection. |
| Conformance inspection: | See MIL-DTL-16377. |

## GENERAL INFORMATION

Assignment of military part numbers and general information is shown in table IV.



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TABLE I. First article and comparison inspection for complete type II fixtures.

| Tests | Requirement Paragraph of MIL-DTL-16377 | Test Paragraph of MIL-DTL-16377 | Remarks |
|---|---|---|---|
| Examination | 3.12 | 4.5 | -- |
| Shock | 3.5.12 | 4.8.7 | -- |
| Vibration | 3.5.13 | 4.8.8 | -- |
| Enclosure effectiveness | 3.5.11 | 4.8.14.2 | Dripproof |
| Continuity of grounding | 3.6.13.4 | 4.8.12 | --- |
| Optical uniformity | 3.7.4.1.1 | 4.8.4 | Symbol 89 only |
| Light output | 3.7.1 | 4.8.6.1 | Candlepower |
| Magnetic permeability | 3.4.8 | 4.8.18 | --- |

TABLE II. First article and comparison inspection for repair parts.

| Tests | Requirement Paragraph of MIL-DTL-16377 | Test Paragraph of MIL-DTL-16377 | Remarks |
|---|---|---|---|
| Examination | 3.12 | 4.5 | --- |
| Optical uniformity | 3.7.4.1.1 | 4.8.4 | 1/ |
| Light output | 3.7.3.7 | 4.8.6.3 | Luminous transmittance 1/ |

NOTE:

1/ For lens of window assembly only.



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TABLE III. First article and comparison inspection for complete type III fixtures.

| Tests | Requirement Paragraph of MIL-DTL-16377 | Test Paragraph of MIL-DTL-16377 | Remarks |
|---|---|---|---|
| Examination | 3.12 | 4.5 | --- |
| Operation | 3.12 | 4.8.1 | --- |
| Power interface | 3.6.18 | 4.8.21 | --- |
| Electromagnetic interference | 3.6.14 | 4.8.16 | --- |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | 1/ |
| Insulation resistance | 3.6.16 | 4.8.3 | 1/ |
| Enclosure effectiveness | 3.5.11 | 4.8.14.4 | 1/ Dripproof, unless otherwise specified. |
| Shock | 3.5.12 | 4.8.7 | 1/ |
| Vibration | 3.5.13 | 4.8.8 | 1/ |
| Enclosure effectiveness | 3.5.11 | 4.8.14.4 | 1/ Dripproof, unless otherwise specified. |
| Dielectric withstanding voltage | 3.6.15 | 4.8.2 | 1/ |
| Insulation resistance | 3.6.16 | 4.8.3 | 1/ |
| Noise | 3.5.14 | 4.8.9 | 1/ |
| Continuity of grounding | 3.6.13.4 | 4.8.12 | 1/ |
| Photometrics | 3.7.1 | 4.8.6 | --- |
| Brightness | 3.7.5 | 4.8.6.4 | --- |
| Chromaticity | 3.7.4 | 4.8.6.5 | --- |
| Correlated color temperature, white light only | 3.7.4.1.1 | 4.8.6.6 | --- |
| Lumen maintenance | 3.7.11 | 4.8.20 | --- |
| Fail-safe circuit design | 3.6.17 | 4.8.22 | --- |
| Optical uniformity | 3.7.6.1.1.1 | 4.8.4 | --- |
| Stress relief | 3.8.5 | 4.8.5 | --- |
| Magnetic permeability | 3.4.8 | 4.8.18 | --- |

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.



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TABLE IV. Part numbers and general information.

| Military Part Number M16377/21- | Item Name | National Stock Number (NSN) | Remarks |
|---|---|---|---|
| 89 | Complete fixture, symbol 89 | 6210-00-780-8502 | Figure 2 |
| 90.2 | Complete fixture, symbol 90.2 | 6210-00-548-0081 | Figure 3 |
| 89SSL | Complete fixture, solid state, symbol 89SSL | --- | --- |
| 90.2SSL | Complete fixture, solid state, symbol 90.2SSL | --- | --- |
| 001 | Window assembly | 6210-00-897-5652 | Figure 4 |
| 002 | Reflector | --- | Figure 6 |
| 003 | Reflector/baffle assembly | 6210-00-683-5472 | Figure 8 |

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-007)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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